

Charge, q (C)

↳ Units, Coulomb (C)

↳ proton = $+e$ $\rightarrow 1.602 \times 10^{-19} \text{ C}$
electron = $-e$
neutron = 0 C

Electromagnetism

↳ presence of charge $\rightarrow$ electric
↳ charge in motion $\rightarrow$ magnetic

↳ Quantized

↳ Conservation of charge: Charge in an isolated system remains constant over time

Electrostatics

Coulomb's Law

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{(d_{12})^2} \vec{a}_{d12} \quad > 0 \text{ when } q_1, q_2 \text{ both -ve or +ve (repulsive)}$$
$$= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{R}_2 - \vec{R}_1|^3} (\vec{R}_2 - \vec{R}_1) \quad < 0 \text{ when } q_1, q_2 \text{ opposite charges (attractive)}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m (permittivity of vacuum)}$$

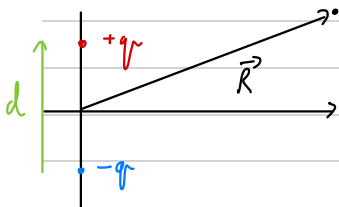
Electrical Field, $\vec{E}$

$$\vec{E}(\rho) \equiv \frac{\vec{F}}{q_{\text{test}}} \left[\frac{N}{C} = \frac{V}{m} \right], q_{\text{test}} \text{ is a small test charge}$$

Point charge at origin

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{R}|^3} \vec{R}$$

Electric dipole



$$\vec{E} = \frac{q}{4\pi\epsilon_0} \left[\frac{\vec{R} - \vec{d}/2}{|\vec{R} - \vec{d}/2|^3} - \frac{\vec{R} + \vec{d}/2}{|\vec{R} + \vec{d}/2|^3} \right]$$

For $d \ll R$

$$\vec{E} = \frac{1}{4\pi\epsilon_0 R^3} \left[3 \frac{\vec{R} \cdot \vec{q} \vec{d}}{R^2} \vec{R} - (q \vec{d}) \right]$$

Electric dipole moment $\rightarrow \vec{p} = q\vec{d}$



Coordinate systems

↳ Unit vectors:

- 1) Unit length
- 2) Point in direction of increasing coordinate
- 3) Normal to corresponding locus of const coordinate

Cartesian

$$\vec{R} = x\vec{a}_x + y\vec{a}_y + z\vec{a}_z$$

Cylindrical

$$\vec{R} = r\vec{a}_r(\phi) + z\vec{a}_z$$

Spherical

$$\vec{R} = \rho \vec{a}_R(\theta, \phi)$$

also $\vec{a}_\theta(\theta, \phi)$

Steps for integrals, flux

- 1.) Pick coordinate system
- 2.) Determine what coordinate varies
- 3.) Select differential element

Field in volume

$$\vec{E} = \int_V \frac{\rho_v dV'}{4\pi\epsilon_0} \frac{(\vec{R} - \vec{R}')}{|\vec{R} - \vec{R}'|^3}$$

$\vec{R}'$ → volume w/ charge in, $\vec{R}$ → Field location

ρ_v = volume density of charge [C/m^3]

Field in Surface

$$\vec{E} = \int_{S'} \frac{\rho_s dS'}{4\pi\epsilon_0} \frac{(\vec{R} - \vec{R}')}{|\vec{R} - \vec{R}'|^3}$$

ρ_s = surface charge density [C/m^2]

Field in line

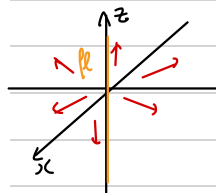
$$\vec{E} = \int_{L'} \frac{\rho_L dL'}{4\pi\epsilon_0} \frac{(\vec{R} - \vec{R}')}{|\vec{R} - \vec{R}'|^3}$$

ρ_L = linear charge density [C/m]

Steps for integration, $\vec{E}$

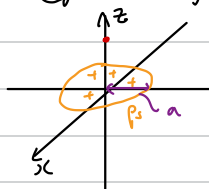
- 1.) Pick coordinate system
- 2.) Write dq' expression → ex. $dq' = \rho_s dS' = \rho_s r' d\phi' dr'$ (for flat plate)
- 3.) Write expressions for $\vec{R}, \vec{R}', \vec{R} - \vec{R}', |\vec{R} - \vec{R}'|$
- 4.) Write expression for $d\vec{E}$
- 5.) Integrate

Infinite line charge



$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \vec{a}_r$$

Charge disk, field @ z axis



$$\vec{E} = \frac{\rho_s a^2}{4\epsilon_0} \left[-\frac{2}{\sqrt{a^2 + z^2}} + \frac{2}{|z|} \right] \vec{a}_z$$

$$\vec{E} = \frac{\rho_s}{2\epsilon_0} \text{ for } z=0$$

$$\vec{E} \rightarrow 0 \text{ for } z \rightarrow \infty$$

Infinite plate

$$\vec{E} = \begin{cases} \frac{\rho_s}{2\epsilon_0} \vec{a}_z, & z > 0 \\ -\frac{\rho_s}{2\epsilon_0} \vec{a}_z, & z < 0 \end{cases}$$

Gauss Law

Integral form

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

Q → total charge in S

Outward flux of $\vec{E}$ through S

Differential form

$$\nabla \cdot \vec{E} = \frac{\rho_v}{\epsilon_0}$$

Sphere of radius a , density ρ_v

$$E_R(R') = \frac{\rho_v}{\epsilon_0} R', \quad R' < a$$

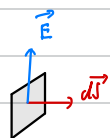
$$= \frac{\rho_v a^3}{\epsilon_0 (R')^2}, \quad R' > a$$

Gaussian surface:

↳ closed

↳ Either $\vec{E} \perp d\vec{S} \rightarrow \vec{E} \cdot d\vec{S} = 0$

or $\vec{E} \parallel d\vec{S}$ & $|\vec{E}|$ constant on $S \rightarrow \oint_S \vec{E} \cdot d\vec{S} = E \int_S dS$



Electrostatic work

$\vec{F}_E$ = force produced by $\vec{E}$ on q

$\vec{F}$ = force required to apply to move the charge

$$\vec{F} = -\vec{F}_E = -q\vec{E}$$

$$W = \int_{\gamma} (-q\vec{E}) \cdot d\vec{L}$$

$$\oint \vec{E} \cdot d\vec{L} = 0$$

$$\oint (-q\vec{E}) \cdot d\vec{L} = 0, \text{ energy conservation } \nabla \times \vec{E} = 0$$

Potential

$$\hookrightarrow \vec{E} = -\nabla V \quad [V] = [J/C]$$

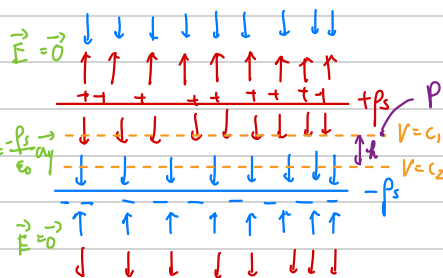
Potential

$$\hookrightarrow \frac{W}{q} = -\int_{P_1}^{P_2} \vec{E} \cdot d\vec{l} = V(P_2) - V(P_1), \text{ defined wrt a ref. point}$$

Convention: If charge distribution is **bounded**
ref point taken at ∞ , i.e.
 $\lim_{R \rightarrow \infty} V(R) = 0$

$\hookrightarrow \vec{E}$ is always normal to surface $V = \text{const}$
always points in direction of steepest decrease of V

Parallel plates

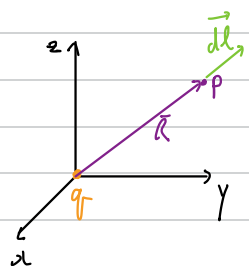


$$V(P) = V(P_2) - V(P_1)$$

$$= 0 - \int_{P_2}^P \vec{E} \cdot d\vec{l}$$

$$= \frac{\rho_s}{\epsilon_0} h$$

Potential from a charge



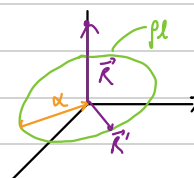
$$V(P) - V(\infty) = - \int_{\infty}^P \vec{E} \cdot d\vec{l}$$

$$V(P) = \frac{q}{4\pi\epsilon_0 R}$$

Superposition potential

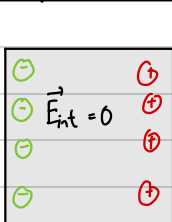
$$V = \int_{V'} \frac{\rho_r dv'}{4\pi\epsilon_0} \frac{1}{|\vec{R} - \vec{R}'|}$$

Charged ring potential on z axis

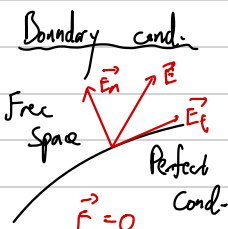


$$V = \frac{\rho_l a}{2\epsilon_0 \sqrt{a^2 + z^2}}$$

Perfect Conductors ($\sigma \rightarrow \infty$)



$$\vec{E} = \vec{E}_{\text{ext}}$$



① $\vec{E} = 0$ inside

② $V = \text{const}$ since $\vec{E} = -\nabla V$

③ $\rho_r = 0$ (volume charge density)

① $E_t = 0$

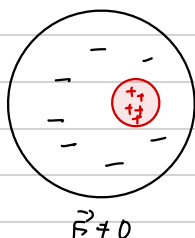
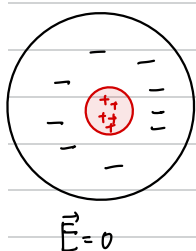
② $E_n = \rho_s / \epsilon_0$

Dielectrics (ideal, $\sigma = 0$)

Ideal case

$\hookrightarrow$ all charges are bound

Non-ideal + dipole moment



Dipole moment

$$\vec{P} = q_n d\vec{h}$$

$$-q_n d\vec{h} \rightarrow +q_n$$

(Polarization vector)

Density of electric dipole moment

$$\vec{P} \equiv \frac{d\vec{p}}{dv} \quad \text{Dipole moment unit volume}$$

Density of polarization charge

$$V = \int_{S'} \frac{\rho_{ps}}{4\pi\epsilon_0 |\vec{R} - \vec{R}'|} dS' + \int_V \frac{\rho_{pv}}{4\pi\epsilon_0 |\vec{R} - \vec{R}'|} dv'$$

ρ_{ps} = surface dist. of charge

ρ_{pv} = volume dist of charge

$$= \vec{P} \cdot d\vec{n}'$$

$$= -\vec{\nabla}' \cdot \vec{P}$$

$$= -(\vec{P}_2 - \vec{P}_1) \cdot d\vec{n}'$$

General Gauss' Law

$$\nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_f \quad \text{free charge} \quad \rho_b, \text{ bound charge}$$

$$\nabla \cdot \vec{D} = \rho_f, \text{ where } \vec{D}, \text{ electric displacement / flux density} \equiv \epsilon_0 \vec{E} + \vec{P}$$

$$\int \vec{D} \cdot d\vec{S} = Q, \text{ integral form}$$

Material dependence

$\hookrightarrow$ Linear, isotropic

$\hookrightarrow$ Electrets

$$\vec{P} = \epsilon_0 \chi_e \vec{E}, \quad \chi_e \text{ [unitless], electric susceptibility}$$

$$\vec{P} = \text{const}$$

$$\vec{D} = \epsilon_0 (1 + \chi_e) \vec{E}$$

$\hookrightarrow$ Piezoelectrics

relative permittivity [F/m]

$$\vec{P} \leftrightarrow \text{mechanical stresses}$$

$$\vec{D} = \epsilon \vec{E}$$

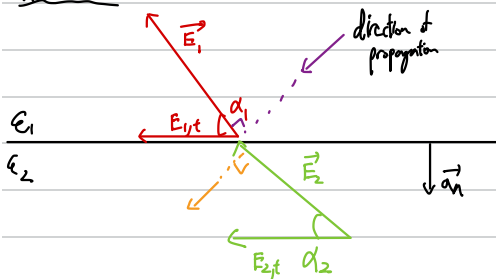
abs permittivity [F/m]

Boundary conds.

$$\hookrightarrow \text{Tangential: } E_{1t} = E_{2t}$$

$$\hookrightarrow \text{Normal: } D_{2n} - D_{1n} = \rho_s \quad (\rho_s = 0 \text{ for ideal dielectrics})$$

Refraction



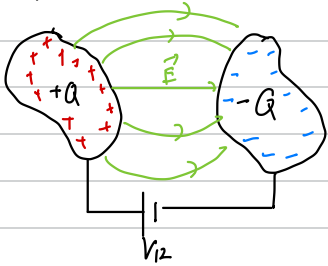
$$\textcircled{1} E_1 \cos \alpha_1 = E_2 \cos \alpha_2 \longrightarrow \epsilon_1 \tan \alpha_1 = \epsilon_2 \tan \alpha_2$$

$$\textcircled{2} \epsilon_1 E_1 \sin \alpha_1 = \epsilon_2 E_2 \sin \alpha_2$$

Dielectric Strength

↳ Max $|\vec{E}|$ material can withstand w/ breaking down

Capacitance



$$Q = C \cdot V$$

↳ capacitance $[F = C/V]$

↳ C depends on geometry, material
NOT on V or Q

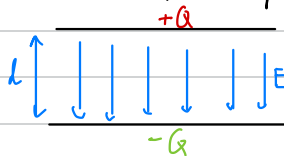
↳ Q proportional to V

i.e. $Q \rightarrow 2Q, \epsilon_s \rightarrow 2\epsilon_s, V \rightarrow 2V$

Method to find capacitance

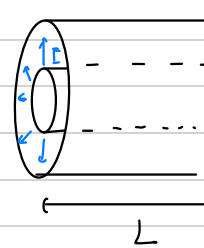
- 1) Choose coord system
- 2) Assume $+Q, -Q$ charge
- 3) Find $\vec{E}$
- 4) Find $V_{12} = -\int \vec{E} \cdot d\vec{l}$
- 5) $C = Q/V_{12}$

Parallel plate capacitor



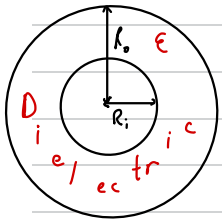
$$C = \epsilon \frac{S}{d}$$

Coaxial cables



$$C = \frac{2\pi \epsilon L}{\ln(b/a)}$$

Conducting sphere



$$C = \frac{4\pi \epsilon}{\frac{1}{R_i} - \frac{1}{R_0}}$$

For isolated sphere, $R_0 \rightarrow \infty$
 $C \rightarrow 4\pi \epsilon R_i$

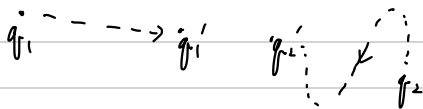
Parallel / Series

$$C_{||} = C_1 + C_2 + \dots + C_n$$

$$\frac{1}{C_{\text{series}}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$

Stored energy

2-charge system



Work needed to bring

$$q_1 \text{ to } \vec{r}_1 \text{ or } q_2 \text{ to } \vec{r}_2: W_c = q_1 V_2(\vec{r}_1) = q_2 V_1(\vec{r}_2)$$

$$= \frac{1}{2} q_2 V_1(\vec{r}_2) + \frac{1}{2} q_1 V_2(\vec{r}_1)$$

General system

$$W_c = \frac{1}{2} \sum_{k=1}^n q_k \cdot V_k$$

potential generated by
all charge except q_k at $\vec{r}_k$

If charge is cont.

$$W_c = \frac{1}{2} \int \rho(\vec{r}') V(\vec{r}') dV'$$

$$= \frac{1}{2} \int_{R^3} \vec{D} \cdot \vec{E} dV'$$

$R^3 \rightarrow \text{entire space}$

$$= \frac{1}{2} \int_{R^3} \epsilon |\vec{E}|^2 dV'$$

Capacitors

$$W = \frac{1}{2} C V^2$$

$$= \frac{1}{2} Q V$$

$$= \frac{1}{2} \frac{Q^2}{C}$$

Force

$$F = \nabla W$$

Poisson Equation

$$\nabla \cdot (\epsilon \nabla V) = -\rho$$

Boundary value problems

$$\begin{cases} \nabla \cdot (\epsilon \nabla V) = -\rho, & \text{in volume } V \\ V|_S = V_0 \end{cases}$$

↳ If $\rho=0$

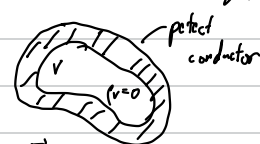
$$\nabla \cdot (\epsilon \nabla V) = 0$$

$$\nabla^2 V = 0 \quad \text{Laplace eq.}$$

Uniqueness

Given ρ, V_0 , solution to Poisson eq. is unique

Electrostatic shielding



$$\vec{E} = 0$$

$$V = \text{const}$$

Not shielding source

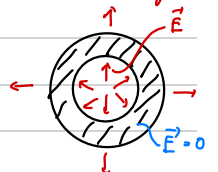
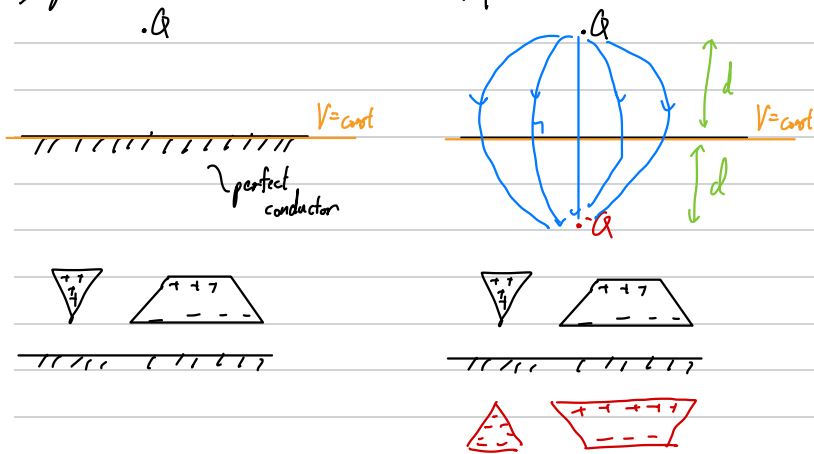


Image Theory

↳ Replace planes with charges

Ex. Original



Current

↳ The motion of charges

↳ Mechanisms

- Conduction
- Electrolysis $\rightarrow$ migration of ions
- Convection
- (Displacement current)

Conduction currents

No $\vec{E} \rightarrow$ free e^- in random motion $\rightarrow$ no "macroscopic" current

$\vec{E} \neq \vec{0} \rightarrow$ Coherent motion of all free e^-
Current

Drift Velocity

$$\vec{u} = -\mu_e \vec{E}$$

(e^- mobility [$\frac{m^2}{Vs}$])

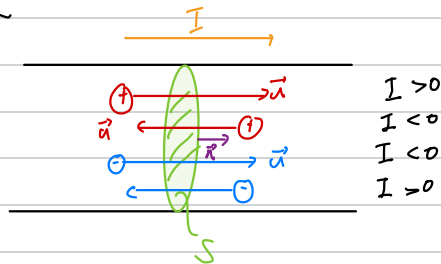
Current intensity (Current) & Current density

$$dI \equiv \frac{dq}{dt}$$

$$I = \int_S dI \quad [A = C/s]$$

$$= \int_S \rho \vec{u} \cdot d\vec{S}$$

$\vec{J} [A/m^2]$, current density



Ohm's Law

Microscopic

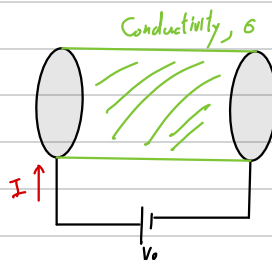
$$\vec{J} = \rho_e \vec{u} = (-\mu_e e) \vec{E}$$

$$= \sigma \vec{E}, \quad \sigma > 0 [S/m], \text{ conductivity}$$

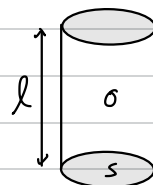
Macroscopic

$$R \equiv V_0 / I$$

Resistance



Resistance of straight object



$$R = \frac{l}{\sigma S}$$

Finding Resistance

- 1) Pick coord system
- 2) Apply voltage V_0

3) Find $\vec{E}$

4) Find $\vec{J} = \sigma \vec{E}$

5) Find $I = \int \vec{J} \cdot d\vec{S}$

6) Find $R = \frac{V_0}{I}$

Charge conservation / Continuity

$$\frac{\partial}{\partial t} \int_V \rho_r(t) dv + \int_S \vec{J} \cdot d\vec{S} = 0 \quad \text{integral form}$$

$$\frac{\partial \rho_r}{\partial t} + \nabla \cdot \vec{J} = 0 \quad \text{differential form}$$

KCL

↳ For static condition,

$$\frac{\partial}{\partial t} \int_V \rho_r(t) dv = 0$$



Then

KVL

Along a closed path,

$$\sum_j V_j = \sum_k R_k I_k$$

sum of emf (voltage rise) = sum of voltage drops across resistances

Relaxation equation (Redistribution of charge in conductor)

Since $\vec{J} = \sigma \vec{E}$

$$\nabla \cdot \vec{J} = \sigma \nabla \cdot \vec{E} = \sigma \frac{\rho_f}{\epsilon} = \frac{-\partial \rho}{\partial t}$$

$$\rho(t) = \rho(0) e^{-t/\tau}, \quad \text{Relaxation time, } \tau = \epsilon/\sigma$$

$$\int_{S_1} \vec{J} \cdot d\vec{S} + \int_{S_2} \vec{J} \cdot d\vec{S} + \int_{S_3} \vec{J} \cdot d\vec{S} = 0$$

Joule's Law

$$P = \int_V \vec{J} \cdot \vec{E} dv$$

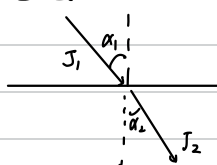
$$= \int_V \sigma |\vec{E}|^2 dv, \quad \text{for } \sigma \neq 0, \text{ lossy material}$$

Current density boundary conditions

$$J_{2n} = J_{1n}$$

$$\frac{J_{2t}}{J_{1t}} = \frac{\sigma_2}{\sigma_1}$$

Conductive media

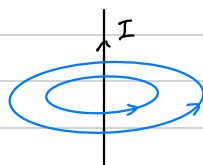
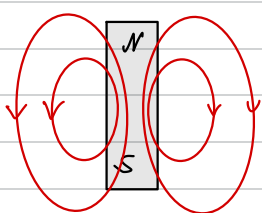


$$\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\sigma_2}{\sigma_1}$$

$$J_2 = J_1 \sqrt{\left(\frac{\sigma_2}{\sigma_1} \sin \alpha_1\right)^2 + \cos^2 \alpha_1}$$

Static magnetic fields

Source:



Magnetic Force, flux density

$$\vec{F}_B = q \vec{u} \times \vec{B}$$

drift velocity

$$\vec{B} \left[\frac{N}{A \cdot m} = T \right], \quad \text{magnetic field or magnetic flux density}$$

Lorentz force eqn.

$$\vec{F} = \underbrace{q \vec{E}}_{\text{electric}} + \underbrace{q \vec{u} \times \vec{B}}_{\text{magnetic}}$$

(microscopic) Magnet
(no magnetic monopoles)

Current (macroscopic)

Postulates of magnetostatics in free space

Differential

$$\nabla \cdot \vec{B} = 0$$

$\Leftrightarrow$

Integral

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

any magnetic flux through closed surface = 0
no magnetic flow sources, lies close upon themselves

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

$\Leftrightarrow$

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I$$

circulation of magnetic flux density in free space, around closed path = $\mu_0(I)$ flux through the surface bounded by the path

(Ampere's Law)

$\hookrightarrow \mu_0$, permeability of free space = $4\pi \times 10^{-7} \text{ H/m}$

$\hookrightarrow 1 \text{ H} = 1 \text{ Tm/A}$

Vector Magnetic Potential

Proportional
 $\nabla^2 \vec{A} = -\mu_0 \vec{J}$ (Poisson eq)

$\vec{A} \left[\frac{\text{Wb}}{\text{m}} \right]$, vector magnetic potential

$\hookrightarrow$ Zero divergence $\vec{B}$ (for particles)

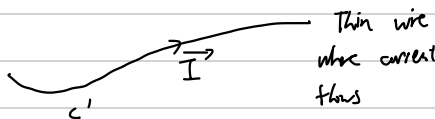
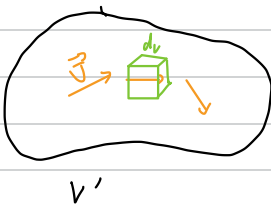
$\vec{A}$ is uniquely defined as:

$$\begin{cases} \vec{B} = \nabla \times \vec{A} \\ \nabla \cdot \vec{A} = 0 \end{cases}$$

Superposition

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\vec{J}(\vec{r}')}{|\vec{R} - \vec{r}'|} dV'$$

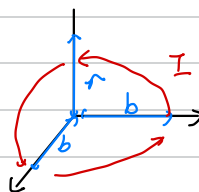
$$\text{or } \vec{A} = \frac{\mu_0 I}{4\pi} \int_{C'} \frac{d\vec{l}'}{|\vec{R} - \vec{r}'|}$$



Biot - Savart Law

$$\vec{B} = \nabla \times \vec{A} = \frac{\mu_0 I}{4\pi} \oint \frac{d\vec{l}' \times (\vec{R} - \vec{r}')}{|\vec{R} - \vec{r}'|^3}$$

Loop $\vec{B}$



$$\hookrightarrow \vec{B} = \frac{\mu_0 I b^2 \vec{a}_z}{2(z^2 + b^2)^{3/2}}$$

Straight wire $\vec{B}$

A diagram of a straight wire of length L carrying current I . A point is located at a distance r from the wire. The magnetic field $\vec{B}$ is shown as a vector pointing out of the page.

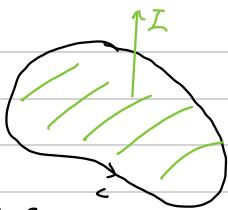
$$\hookrightarrow \vec{B} = \frac{\mu_0 I}{4\pi} \left(\frac{\vec{a}_\phi}{r} \cdot \frac{2L}{\sqrt{r^2 + L^2}} \right)$$

as $\vec{L} = \frac{\mu_0 I}{2\pi r} \vec{a}_\phi$

Ampere's Law

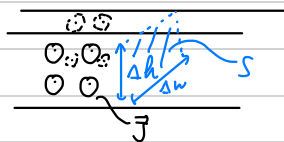
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

↳ For any closed path c



Surface Current Density, $\vec{J}_s$

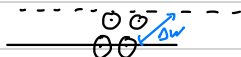
Normal Current density, J



$$I = J \Delta w \Delta h$$

$[A/m^2]$

Surface Current density, J_s



$$I = J_s \cdot \Delta w$$

$[A/m]$

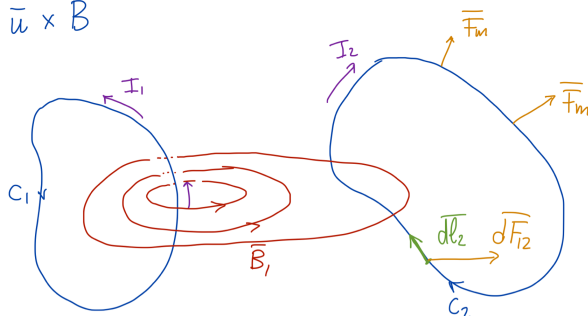
Ampere's Force Law

Current I_1 flowing in c_1 will produce

a.) Magnetic flux density $\vec{B}_1$

b.) Magnetic force on c_2

$$\vec{F}_m = q \vec{u} \times \vec{B}$$



$$d\vec{F}_{12} = dq_2 \vec{u}_2 \times \vec{B}_1$$

$$= dq_2 d\vec{l}_2 \times \vec{B}_1$$

$$= I_2 d\vec{l}_2 \times \vec{B}_1$$

$$\vec{F}_{12} = \int_{c_2} I_2 d\vec{l}_2 \times \vec{B}_1$$

Magnetic Dipole

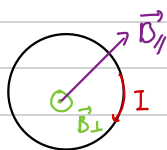
↳ Loop carrying a current



Magnetic Dipole Moment, $\vec{m} = I \cdot S \vec{a}_n$

$|\vec{B}| \sim |\vec{m}|$

Torque



$\vec{D}_1$:

$$d\vec{F}_n = I d\vec{l} \times \vec{B}$$

$$\vec{F}_n = 0 \text{ (No net force)}$$

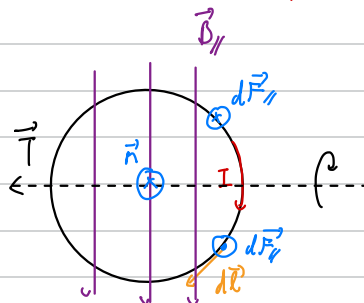
$$\vec{\tau} = 0 \text{ (No torque)}$$



$\vec{D}_2$:

$$\vec{F}_n = 0$$

$$\text{But } \vec{\tau} = \vec{m} \times \vec{B} \neq 0$$



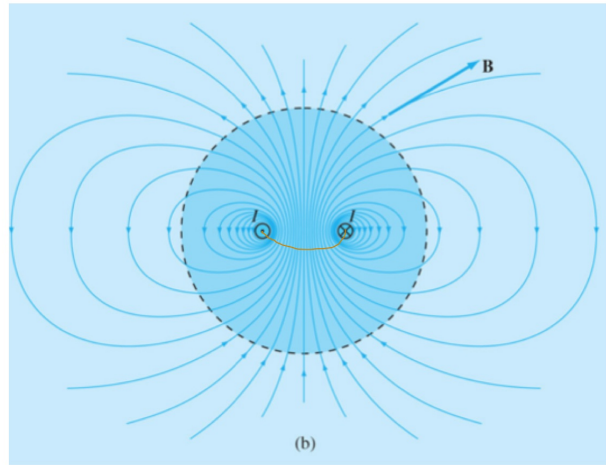
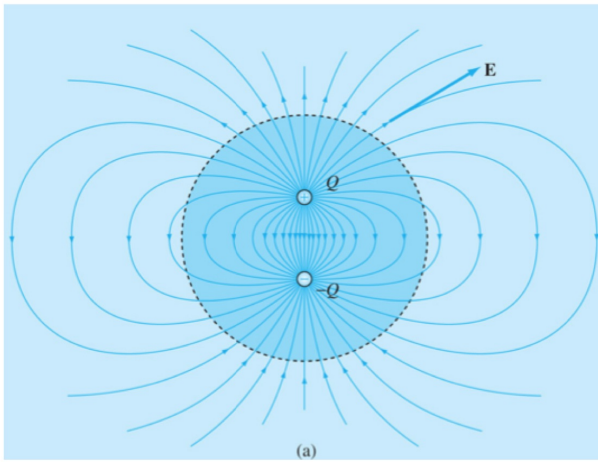
Note:

$\otimes \rightarrow$ into page

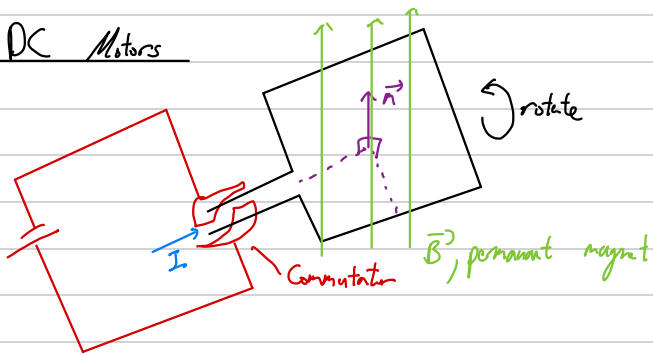
$\odot \rightarrow$ out of page

Electric field of an electric dipole

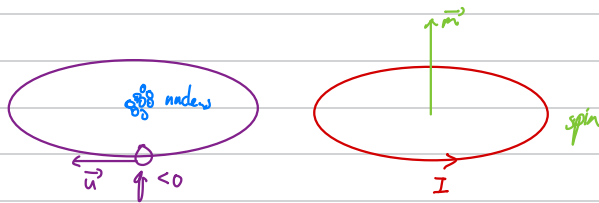
Magnetic flux of a magnetic dipole



DC Motors

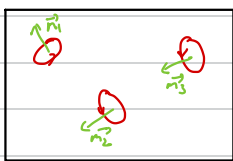


Materials in magnetic field



No external field, $B_{ext} = 0$

$B_{ext} \neq 0$



random orientation



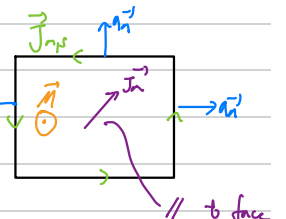
B_{ext}

$\sum_i \vec{B}_i \approx 0$, field due to atoms is

★ Magnetization vector, $\vec{M} = \sum_{heavy} \vec{m}_i / \Delta v$ (density of magnetic dipole moment)

★ Magnetization current density

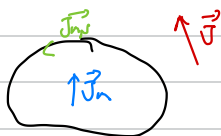
Surface, $\vec{J}_{ms} = \vec{M} \times \vec{a}_n$
Volume, $\vec{J}_m = \nabla \times \vec{M}$



Generalized Ampere's Law

Free Space

Generic material



$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

$$\nabla \times \vec{B} = \mu_0 (\vec{J} + \vec{J}_n)$$

$$\& \vec{J}_n = \nabla \times \vec{M}$$

$$\nabla \times \left[\frac{\vec{B}}{\mu_0} - \vec{J} \right] = \vec{J} \rightarrow \text{Current density of free charges} \quad \text{or} \quad \oint \vec{H} \cdot d\vec{l} = I$$

$\vec{H}$, Magnetic field [A/m]



Permanent magnets

Nonlinear magnets

Linear materials

$$\vec{N} = \text{const}$$

$$\vec{A} = f(\vec{H})$$

$$\hookrightarrow \vec{M} = \chi_m \vec{H}$$

Magnetic susceptibility

↳ Then $\vec{B} = \mu_0 (1 + \chi_m) \vec{H}$

Relative permeability $\mu_r = 1 + K_m$

Absolute permeability, $\mu = \mu_0 \mu_r = \mu_0 (1 + \chi_m)$

Magnetic Materials

Diamagnetic materials

$$\hookrightarrow \vec{A} = 0 \quad \text{when} \quad \vec{H} = 0$$

↳ When $\vec{H} \neq 0$, $\vec{M} \approx 0$

↳ $\mu_r \leq 1$

ex. Binomial $\mu_r = 0.9999$

- Superconductor

When $T < T_c$,

$$X_n = -1, \mu_n = 1 + X_n = 0, \vec{B} = 0$$

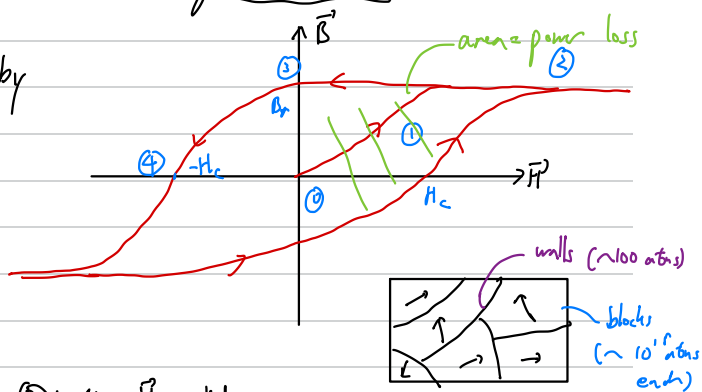


Paramagnetie materiaal

↳ Magnetizer disturbed by thermal vibrations

ex. Al $\mu_r = 1.00002$

Ferromagnetic materials



⑥: No F applied

①: $\vec{H}$ applied, domains aligned to $\vec{H}$ grows,
the stress

②: Saturated

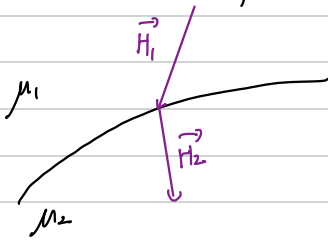
③: $T < \text{Curie temperature}$, a magnet!

④: H_c : Coercive field intensity (intensity for demagnetization)

↳ Hard materials: $H_c > 10^5$ A/m ex. magnets

↳ Soft materials: $H_c < 10^5 \text{ A/m}$ ex. Transformers, AC machines

Magnetic Boundary conditions

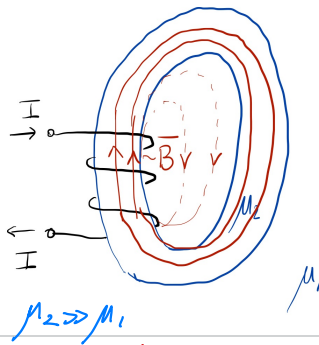
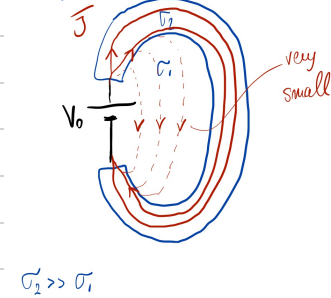


Normal: $B_{1n} = B_{2n}$

Tangential: $\vec{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{J}$

Electrical vs Magnetic circuits

steady state (DC)



Electrical

$$\begin{cases} \nabla \cdot \vec{J} = 0 \text{ (steady)} \\ \nabla \times \vec{E} = 0 \rightarrow \nabla \times \frac{\vec{J}}{\sigma} = 0 \end{cases}$$

Magnetic

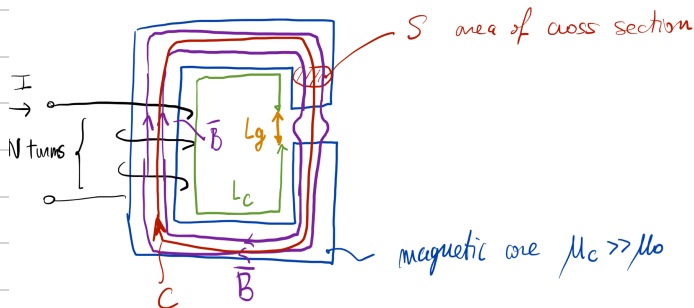
$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{H} = 0 \rightarrow \nabla \times \frac{\vec{B}}{\mu} = 0 \end{cases}$$

(Boites in wire)

$$\vec{J} \Leftrightarrow \vec{B}$$

$$\sigma \Leftrightarrow \mu$$

Magnetic Circuit



Assumptions:

- 1) All $\vec{B}$ lines confined into core
- 2) Short gaps $\vec{E} \perp \vec{B}$
- 3) Small area $\vec{J}_{\text{max}}$, $\Phi_m = \int \vec{B} \cdot d\vec{s} \approx BS$

a) Start with Ampere's Law

$$\oint \vec{H} \cdot d\vec{l} = NI$$

$$\oint \frac{\vec{B}}{\mu} \cdot d\vec{l} + \int \frac{\vec{B}}{\mu_0} d\vec{l} = NI$$

c) Substitute (b)

$$\int \frac{\Phi_m}{\mu_c S} dl + \int \frac{\Phi_m}{\mu_0 S} dl = NI$$

$$\frac{\Phi_m L_c}{\mu_c S} + \frac{\Phi_m L_g}{\mu_0 S} = NI$$

b) Since $\vec{B} \parallel d\vec{l}$,

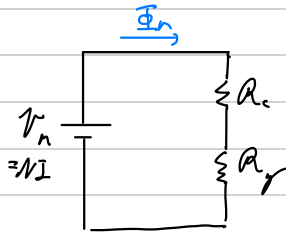
$$\vec{B} \cdot d\vec{l} = B dl$$

$$= \frac{\Phi_m}{S} dl$$

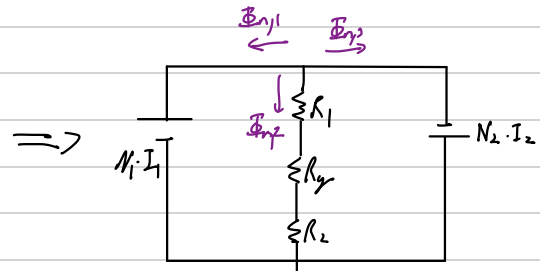
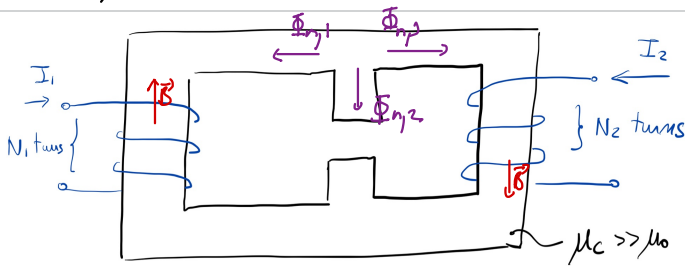
$$d) \Phi_m = \frac{NI}{\frac{L_c}{\mu_c S} + \frac{L_g}{\mu_0 S}}$$

Summary

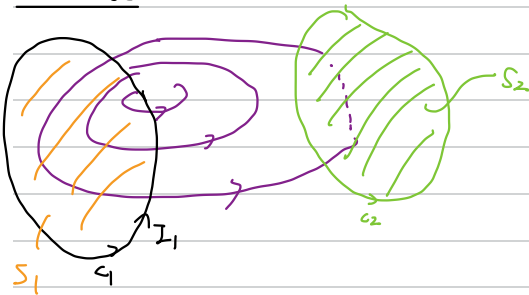
Elec. Circuit	Mag. Circuit
Voltage (electromotive force)	$\mathcal{V}_m = N \cdot I$ (magnetomotive force)
Current	Φ_m Magnetic flux
Resistance, R	$\mathcal{R}$ Reluctance <i>cursive</i>



Frequency circuit



Inductance



Magnetic flux through S_2 caused by I_1 :

$$\Phi_{12} = \int_{S_2} \vec{B}_1 \cdot d\vec{S}_2$$

$$\rightarrow \Phi_{12} \propto I_1$$

$$= L_{12} \cdot I_1 \quad \left[H = \frac{Wb}{A} = \frac{Vs}{A} = \frac{Tm^2}{A} \right]$$

$L_{12} \rightarrow$ Mutual inductance for S_1, S_2

Flux through S_1 :

$$\Phi_{11} = \int_{S_1} \vec{B}_1 \cdot d\vec{S}_1$$

$$= L_{11} \cdot I_1 \quad \text{--- self ---}$$

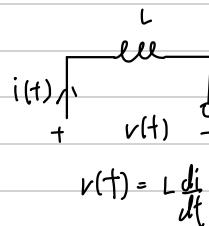
(Self inductance)

Inductance properties

$\hookrightarrow$ Depends on geometry

$\hookrightarrow$ Depends on μ

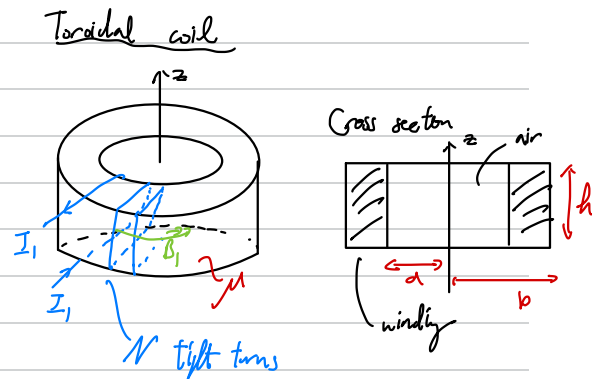
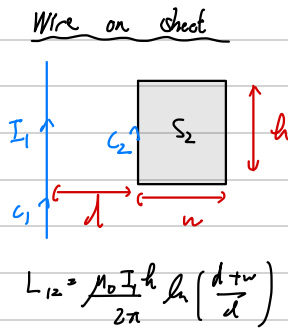
$\hookrightarrow$ Depends on I for non-linear materials



$\hookrightarrow \Phi_{11} \geq \Phi_{12} \Rightarrow L_{11} \geq L_{12}$ always true

Steps in inductance calculation (Method 1)

- 1) Pick coord. system
- 2) Assume I_1
- 3) Find $\vec{B}_1$
- 4) Find $\vec{E}$
- 5) $L = \frac{\Phi}{I}$



$$\vec{B}_1 = \vec{a}_\phi \frac{\mu N I_1}{2\pi r}$$

$$\hookrightarrow \Phi_w = \frac{\mu N I_1}{2\pi} h \ln\left(\frac{b}{a}\right) \quad \text{flux of 1 winding}$$

$$\hookrightarrow \Phi_m = N \Phi_w = \frac{\mu N^2 I_1}{2\pi} h \ln\left(\frac{b}{a}\right) \quad \text{total magnetic flux}$$

$$\hookrightarrow L = \frac{\mu N^2}{2\pi} h \ln\left(\frac{b}{a}\right)$$

Magnetic energy

$$W_m = \frac{1}{2} \int_{R^3} \vec{B} \cdot \vec{H} dV$$

In induction

$$I \rightarrow \text{---} \text{---} \text{---} \text{---} \quad W_m = \frac{1}{2} L I^2$$

Steps in inductance calculation (Method 2)

- 1) Coord system
- 2) Assume current I
- 3) find $\vec{B}$
- 4) find $W_m = \frac{1}{2} \int \vec{B} \cdot \vec{H} dV$
- 5) $L = \frac{2W_m}{I^2}$

Coaxial Cable

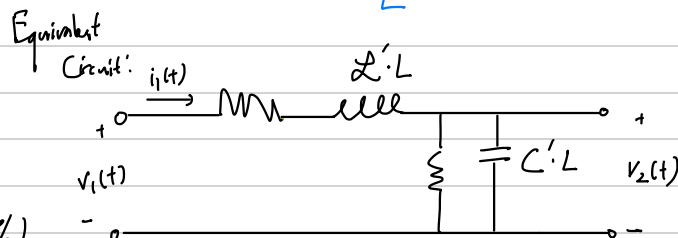
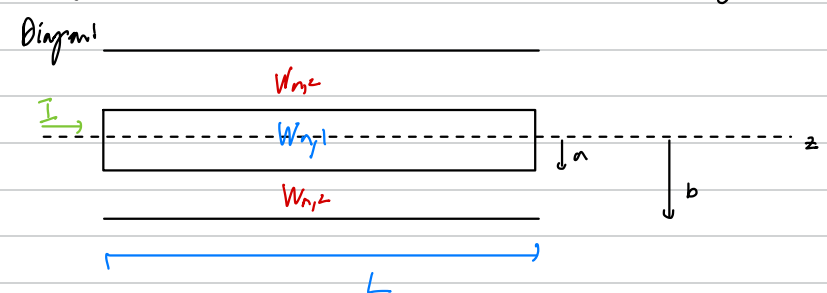
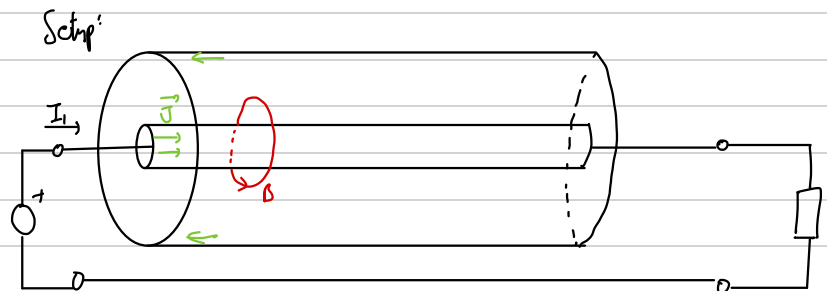
$$\hookrightarrow \vec{B} = \begin{cases} \frac{\mu_0 I}{2\pi a} & r < a \\ \frac{\mu_0 I}{2\pi r} & a < r < b \\ 0 & r > b \end{cases}$$

$$\hookrightarrow W_{m1} = \frac{\mu_0 I^2 L}{16\pi}$$

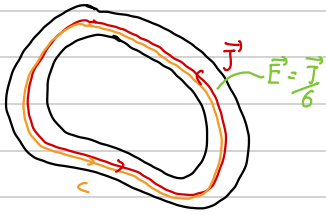
$$\hookrightarrow W_{m2} = \frac{\mu_0 I^2 L}{4\pi} \ln\left(\frac{b}{a}\right)$$

$$\hookrightarrow W_m' / L = \frac{\mu_0 I^2}{16\pi} + \frac{\mu_0 I^2}{4\pi} \ln\left(\frac{b}{a}\right)$$

$$\mathcal{L}, \text{ Inductance per unit length} = \frac{2W_m'}{I^2} = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$$



Non-conservative $\vec{E}$



Sources of non-conservative $\vec{E}$

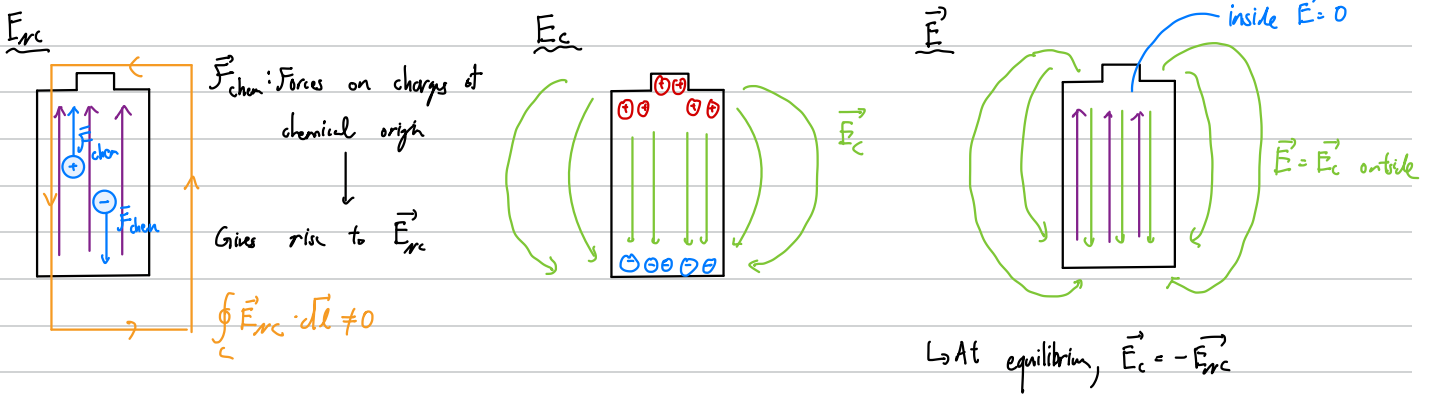
- ↳ Chemical batteries
- ↳ Photovoltaic cells
- ↳ Induction

↳ Changing current $\rightarrow$ non-conservative $\vec{E}$ field

i.e. $\oint_C \vec{E} \cdot d\vec{l} \neq 0$

Chemical battery

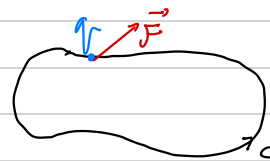
$$\vec{E} = \underbrace{\vec{E}_c}_{\text{conservative part}} + \underbrace{\vec{E}_{nc}}_{\text{Non-conservative part}}$$



Electromotive force

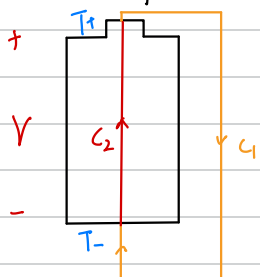
$$V_{\text{emf}} \triangleq \oint_C \vec{E} \cdot d\vec{l}$$

Work done by field in closed path
 normalized



$$= \oint_C (\vec{E} + \vec{u} \times \vec{B}) \cdot d\vec{l}$$

In battery



$$\begin{aligned} V_{\text{emf}} &= \oint_{C_1+C_2} \vec{E} \cdot d\vec{l} \\ &= \oint_{C_1+C_2} \vec{E}_{nc} \cdot d\vec{l} + \int_{C_1+C_2} \vec{E}_c \cdot d\vec{l} \\ &= \int_{C_2} \vec{E}_{nc} \cdot d\vec{l} = \int_{C_1} \vec{E}_c \cdot d\vec{l} \\ &= - \int_{T_1}^{T_2} \vec{E}_c \cdot d\vec{l} \end{aligned}$$

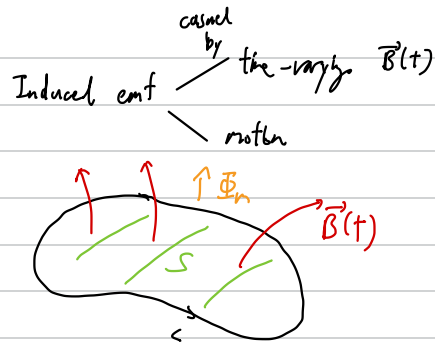
Faraday's Law

Integral form

$$\oint_C \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int_S \vec{B} \cdot d\vec{S}$$

$\mathcal{V}_{\text{emf}}$: electromotive force

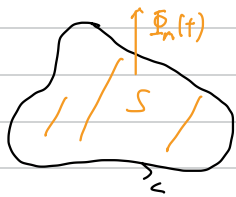
$\Phi_m(t)$: magnetic flux through S



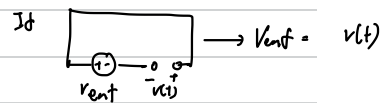
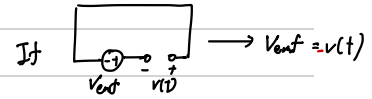
Differential form

$$\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

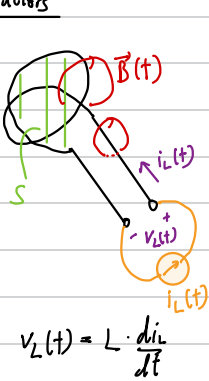
EMF as voltage



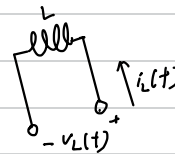
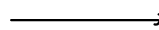
Right Hand Rule



Inductors

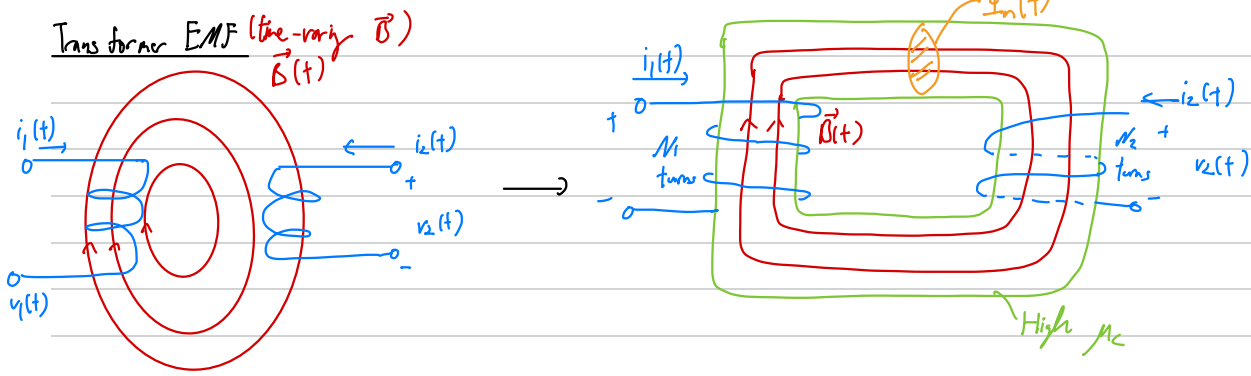


Equivalent circuit

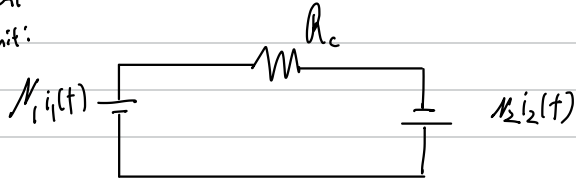


$$\mathcal{V}_L(t) = - \mathcal{V}_{\text{emf}}(t) = L \frac{di_L}{dt}$$

- ① $i_L(t)$ produces $\vec{B}(t)$
- ② $\vec{B}(t)$ produces $\wedge$ flux $\Phi_m(t) = L \cdot i_L(t)$
- ③ EMF induced $\mathcal{V}_{\text{emf}}(t) = - \frac{d}{dt} \Phi_m(t) = -L \frac{di_L}{dt}$

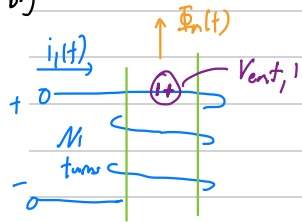


a) Equivalent Circuit:



$$\Phi_m(t) = \frac{N_1 i_1(t) + N_2 i_2(t)}{R_c}, \quad R_c = \frac{l_c}{\mu_c \cdot S}$$

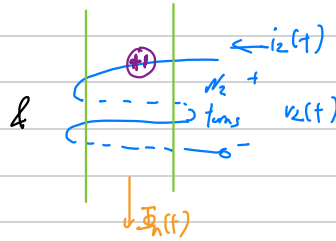
b)



Apply Faraday's Law,

$$V_{emf,1} = -N_1 \frac{d\Phi_m}{dt}$$

$$v_1(t) = -V_{emf,1} = N_1 \frac{d\Phi_m}{dt}$$



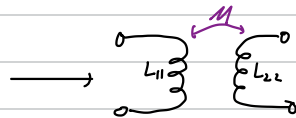
$$V_{emf,2} = -N_2 \frac{d\Phi_m}{dt}$$

$$v_2(t) = N_2 \frac{d\Phi_m}{dt}$$

Thus, $\frac{v_2(t)}{v_1(t)} = \frac{N_2}{N_1}$

c) Combine a, b

$$v_1(t) = \frac{N_1^2}{R_c} \frac{di_1}{dt} + \frac{N_1 N_2}{R_c} \frac{di_2}{dt}$$

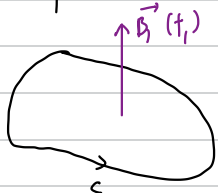


$$v_2(t) = \frac{N_1 N_2}{R_c} \frac{di_1}{dt} + \frac{N_2^2}{R_c} \frac{di_2}{dt}$$

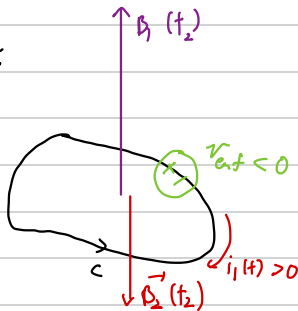
Here, $M = \sqrt{L_{11} L_{22}}$

Lenz Law

At t_1 :



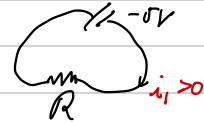
At t_2 :



$$\Phi_m(t_2) > \Phi_m(t_1)$$

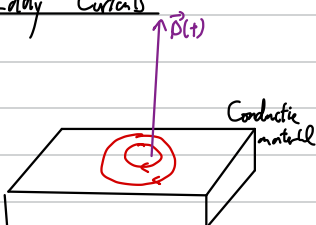
$$V_{emf} = -\frac{d\Phi_m}{dt} < 0$$

Equivalent Circuit:



→ Current in direction that opposes field change

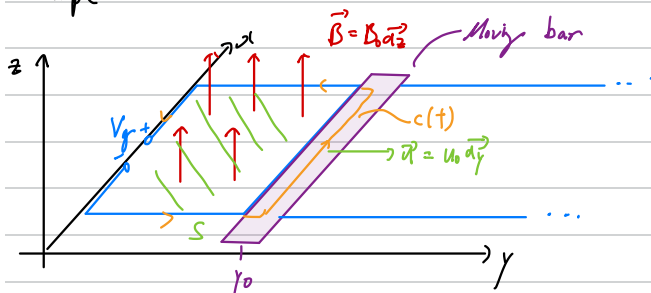
Eddy Currents



$$\vec{j} \perp \vec{B}$$

Motional EMF

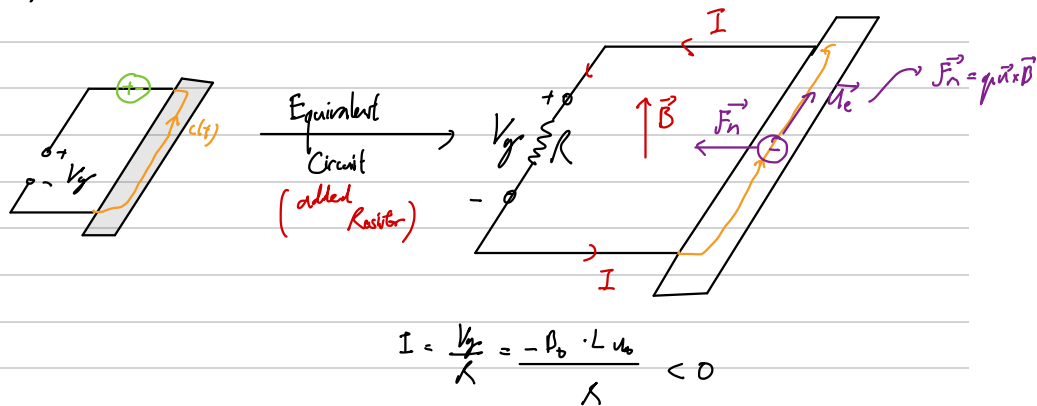
Example



$$S(t) = L(y_0 + u_0 t)$$

$$\Phi_m(t) = \int \vec{B} \cdot d\vec{S} = B_0 \cdot L(y_0 + u_0 t)$$

$$V_{ind} = -\frac{d}{dt} \Phi_m = -B_0 \cdot L u_0 = V_g$$

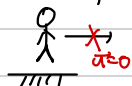


Reference Frame

$$\vec{F} = q\vec{E} + q\vec{u} \times \vec{B}$$

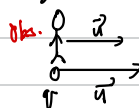
electric force magnetic force

Stationary:



$$\vec{F} = q\vec{E} + q\vec{u} \times \vec{B}$$

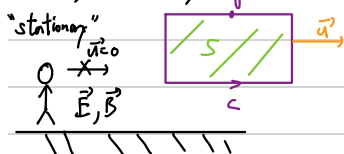
Moving with charge:



$$\vec{F} = q\vec{E}'$$

$$\text{Then } \vec{E} + \vec{u} \times \vec{B} = \vec{E}'$$

Total, Transformer, Motional EMF



Stokes' Theorem

$$V_{ind} = \oint \vec{F} \cdot d\vec{l} = \oint \vec{E} \cdot d\vec{l} + \oint (\vec{u} \times \vec{B}) \cdot d\vec{l} = \int \nabla \times \vec{E} \cdot d\vec{S} + \oint (\vec{u} \times \vec{B}) \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{S} + \oint (\vec{u} \times \vec{B}) \cdot d\vec{l}$$

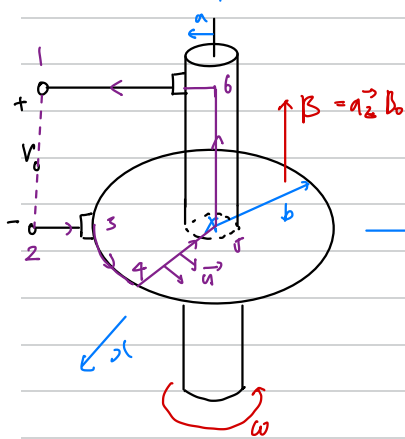
$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$

Faraday's Law, Differential form

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$

$$V_{ind} = \underbrace{-\frac{d}{dt} \int \vec{B} \cdot d\vec{S}}_{\text{total emf}} = \underbrace{-\int \frac{d\vec{B}}{dt} \cdot d\vec{S}}_{\text{transformer emf}} + \underbrace{\oint (\vec{u} \times \vec{B}) \cdot d\vec{l}}_{\text{motional emf}}$$

Faraday's disk



$$V_{\text{emf}} = \oint (\vec{u} \times \vec{B}) \cdot d\vec{\ell} \quad , \quad \vec{u} = \omega r \vec{a}_\phi \quad (\text{from } 4 \rightarrow 5)$$

$$= \int_4^5 (\vec{u} \times \vec{B}) \cdot d\vec{\ell} = -\omega B_0 \left(\frac{b^2 - a^2}{2} \right)$$

4 → 5:

$$\vec{u} \times \vec{B} \parallel d\vec{\ell}$$

$$\hookrightarrow \vec{u} \times \vec{B} \cdot d\vec{\ell} \neq 0$$

5 → 6 or 1 → 2:

$$\vec{u} \times \vec{B} \perp d\vec{\ell}$$

$$\hookrightarrow \vec{u} \times \vec{B} \cdot d\vec{\ell} = 0$$

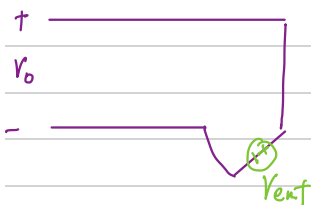
3 → 4:

$$\vec{u} \times \vec{B} \perp d\vec{\ell}$$

$$\hookrightarrow \vec{u} \times \vec{B} \cdot d\vec{\ell} = 0$$

Only portions of $\vec{B}$ lines because of motion contribute to emf

Equivalent circuit



$$V_0 = V_{\text{emf}} = -\omega B_0 \left(\frac{b^2 - a^2}{2} \right)$$

Ampere-Maxwell Law

Differential form:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Current density $\vec{J}$ Displacement current density $\frac{\partial \vec{D}}{\partial t}$

Displacement Current Density

Deformation of e^- clouds $\rightarrow$ motion of charges

$$\vec{J}_D = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \frac{\partial \vec{P}}{\partial t} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

not associated to any motion of charge

Integral form:

$$\oint \vec{H} \cdot d\vec{\ell} = \int \vec{J} \cdot d\vec{s} + \int \frac{\partial \vec{D}}{\partial t} \cdot d\vec{s}$$

Maxwell's Equations

Differential form

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Integral form

$$\oint \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{s} = -\frac{d\phi}{dt}$$

Law

Faraday's Law

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\oint \vec{H} \cdot d\vec{\ell} = \int \vec{J} \cdot d\vec{s} + \frac{d}{dt} \int \vec{D} \cdot d\vec{s}$$

Ampere's Law

$$\nabla \cdot \vec{D} = \rho$$

$$\oint \vec{D} \cdot d\vec{s} = Q$$

Gauss' Law

$$\nabla \cdot \vec{B} = 0$$

$$\oint \vec{B} \cdot d\vec{s} = 0$$

Gauss's Law for magnetostatics (no isolated magnetic charges)